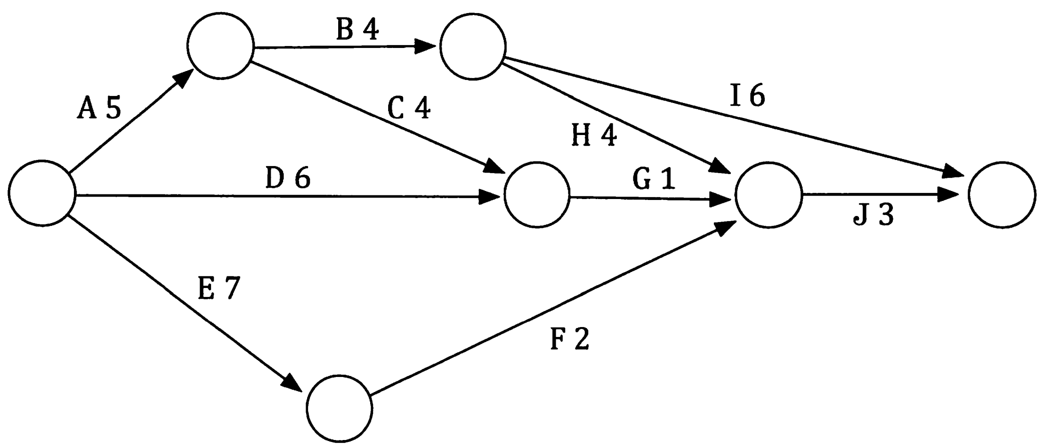
# Chapter 7.

# Project networks

## Situation

A company is contracted to carry out a particular job. The company splits the job into ten smaller tasks and sub-contracts each of these to other companies and individuals. The main company then monitors the progress of the job by attempting to keep it to the program shown in the network below.




- In this network the ten tasks are labelled A to J and are represented by the directed edges, or arcs. The number on each arc shows the time in days that is allowed for the task.
- The network indicates any task that must be completed before another can commence. Thus task H, for example, has 4 days allowed for it and requires task B to be completed before it can be commenced (and task B required A to be completed before it could be commenced). Provided these "prior task commitments" are met other tasks can be going on at the same time. For example tasks A, D and E can all be going on at once.

What is the least number of days required for the job to be completed according to the time allocations on the above schedule?

Poor weather may cause task C to take 6 days rather than 4. If this occurs what will be the least number of days for the job now?

If task I were to be completed in 5 days rather than 6, and all others were to be completed using the allotted time, what will be the least number of days for the job now?

If task J were to be completed in 1 day rather than 3, and all others were to be completed using the allotted time, what will be the least number of days for the job now?

**Project networks.**

The situation on the previous page involved a project network. How did you get on? Did you develop a systematic approach? Did you manage to think it through and answer the questions or did all the things you had to consider leave you confused?

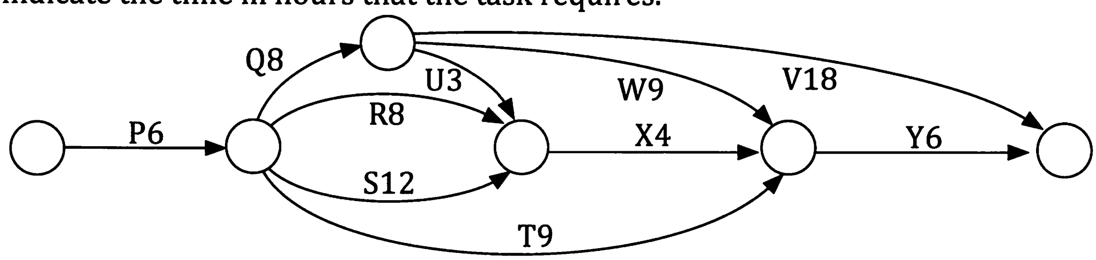
One systematic approach involves two stages:

1. **Forward scan.** I.e. follow the network in the direction of the arrows and at each vertex record the least time needed to reach this stage in the project with all prior tasks completed. In this way we consider the **earliest start time** (EST) of an activity.
2. **Backward scan.** I.e. backtrack through the network recording the latest time each vertex could be reached without delaying the minimum completion time. In this way we consider the **latest start time** (LST) of an activity.

This approach is demonstrated in the following example.

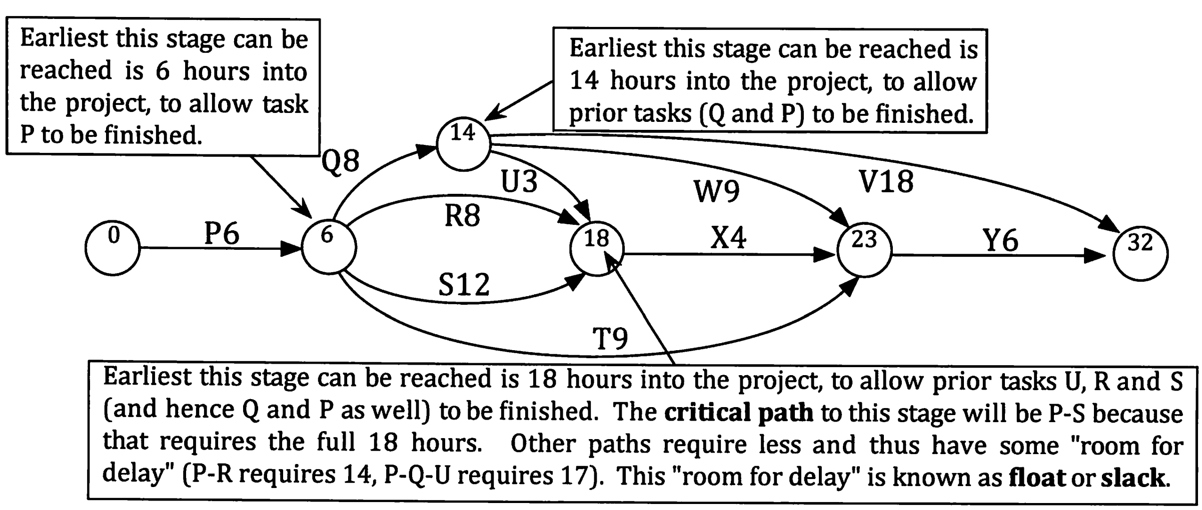
**Example 1**

Find the **minimum completion time** and the **critical path** for the project network shown below. The arcs P, Q, ... , Y represent the tasks involved and the numbers on the arcs indicate the time in hours that the task requires.



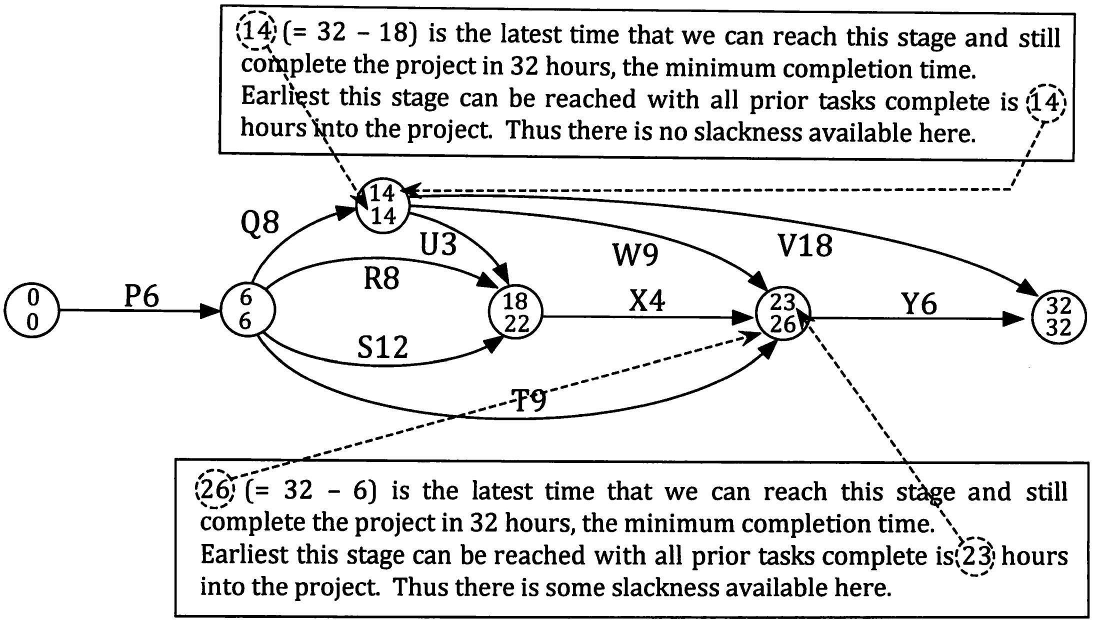

What is the maximum number of extra hours that could be allocated to task S, over and above the 12 in the network, without extending the minimum completion time?

Work through the network from the start and at each vertex record the least time needed to reach that stage in the project with all prior tasks completed. Check that you understand the placement of the numbers in the circles below.




Thus the **minimum completion time** is 32 hours.

Now, to identify the slackness in the system, work back through the network and at each vertex record in the lower half of the circle the latest time we could reach that stage whilst still being able to complete the project in the minimum completion time. Check that you understand the placement of these numbers in the bottom halves of the circles below.




Any vertex for which the earliest time the vertex can be reached is the same as the latest time the vertex can be reached lies on the **critical path**. If we journey from start to finish, through such vertices we can obtain the critical path, P - Q - V. If any of these tasks are delayed the minimum completion time will be extended.

Consider task S and notice that this task will be ready to start 6 hours into the project and does not have to finish until 22 hours into the project. Thus we could allow 16 hours for this 12 hour task. Thus we could allocate an extra 4 hours for task S, over and above the 12 in the network. Task S is said to have a **float**, or **slack** time of 4 hours.

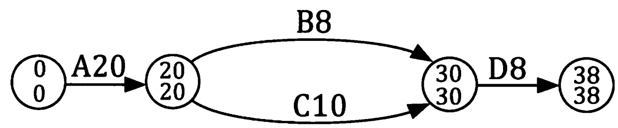
Consider task X. We say that X has an **earliest start time** (EST) of 18 hours and a **latest start time** (LST) of 22 hours (26 – 4). X has 4 hours float.

Consider task W. It has an **earliest start time** (EST) of 14 hours and a **latest start time** (LST) of 17 hours (26 – 9). W has 3 hours float.

**All activities on the critical path have zero float time.**

Note:
- In the above example the diagram is drawn twice and comments are included in boxes. The reader would not have to include such comments and would only need one diagram.
- The float, or slack, considered here is the time by which an activity can be delayed, from its early start time, without altering the project's minimum completion time. It is sometimes referred to as the **total float** of an activity.

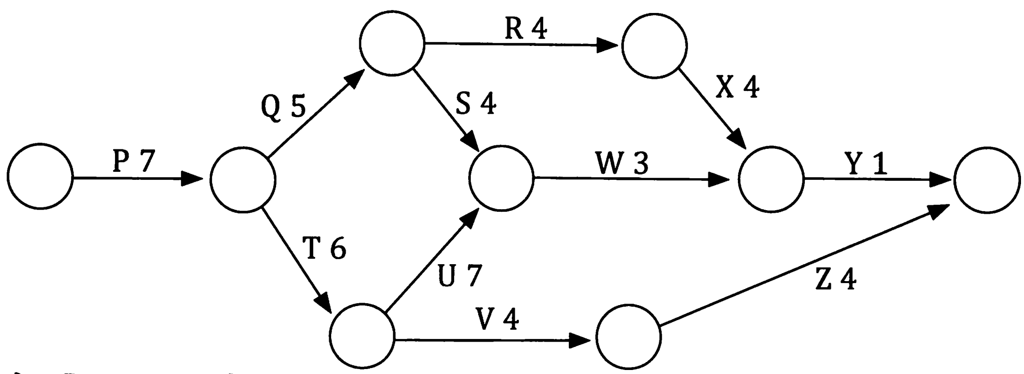
Note: • Consider the diagram:




The numbers in the circles may at first glance give the impression that there is no float, or slack, in the system but with two arcs from the second vertex to the third vertex the critical path is determined by taking task C, the task that takes the greater time. Thus A - C - D is the critical path. Task B has 2 units of float, a fact that can be missed if we only look at the numbers in the circles.

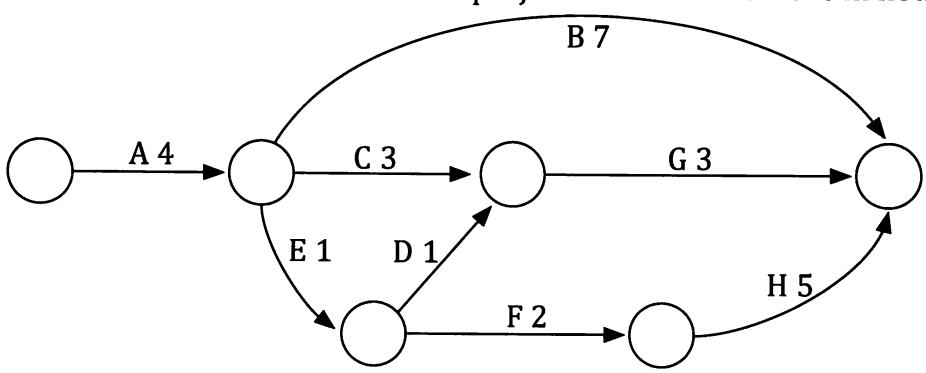
**Exercise 7A**

1. The times for tasks P to Z shown in the project network below are in days.



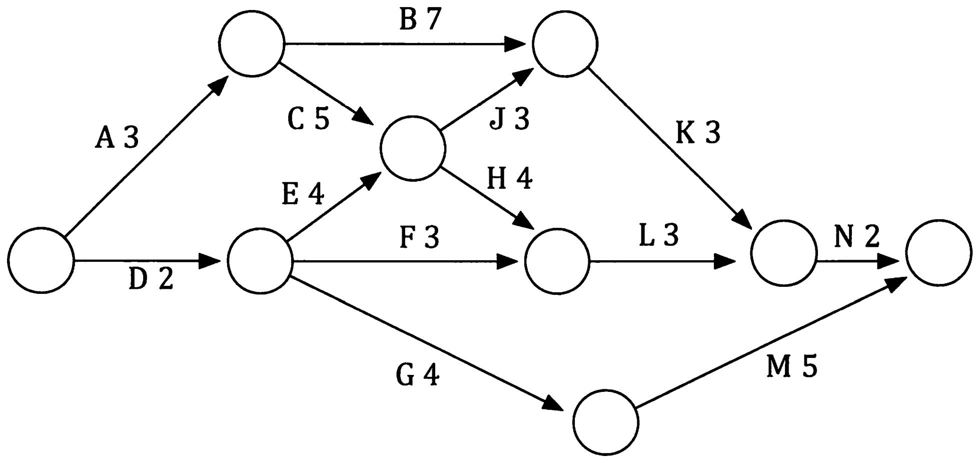

   (a) Determine the minimum completion time.
   (b) Determine the critical path.
   (c) Task V is a task that is expected to take 4 days. Assuming all the other tasks can be completed in exactly the stated times, how many extra days could be allocated to task V without altering the minimum completion time?

2. The times for tasks A to H shown in the project network below are in hours.



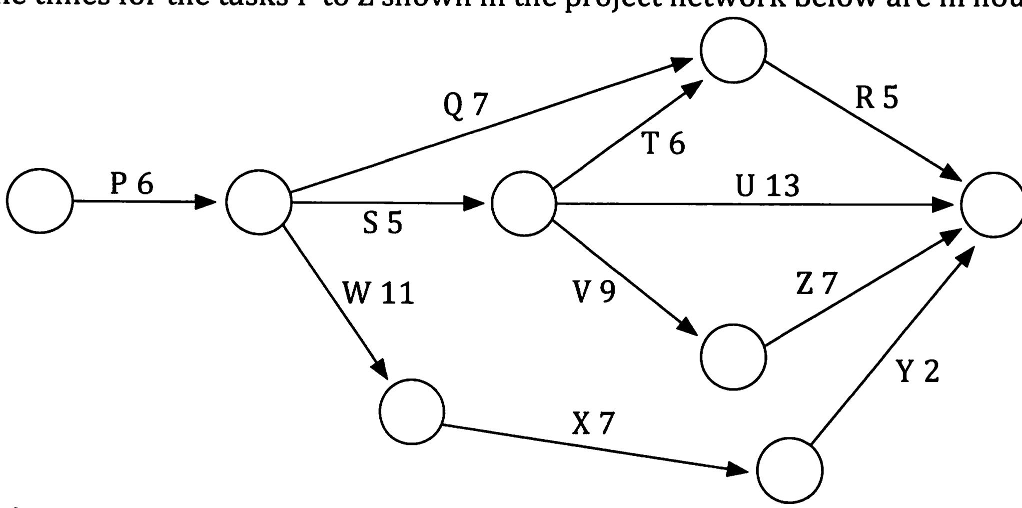

   (a) Determine the minimum completion time.
   (b) Determine the critical path.
   (c) Some problems are experienced with task D and it requires an extra 3 hours. What is the minimum completion time for the project now?
   (d) How many extra hours could be allowed for task B, over and above the 7 hours allocated on the above network, without extending the minimum completion time?

3. The times for the tasks shown in the project network below are in days.




(a) Determine the minimum completion time.
(b) Determine the critical path.
(c) Due to poor weather, tasks D and B each suffer a two day delay in commencement. What is the minimum completion time now?

4. The times for the tasks P to Z shown in the project network below are in hours.




(a) Determine the minimum completion time.
(b) Determine the critical path.
(c) The project managers can employ extra staff and reduce the time required for task V by 3 hours. If they take this option will the minimum completion time be cut by 3 hours?
(d) Instead of the option outlined in part (c) the managers decide to allocate extra staff to <u>one</u> particular task and this reduces the time required for that task by 2 hours and it cuts the minimum completion time by 2 hours. Which task did they allocate the extra staff to?

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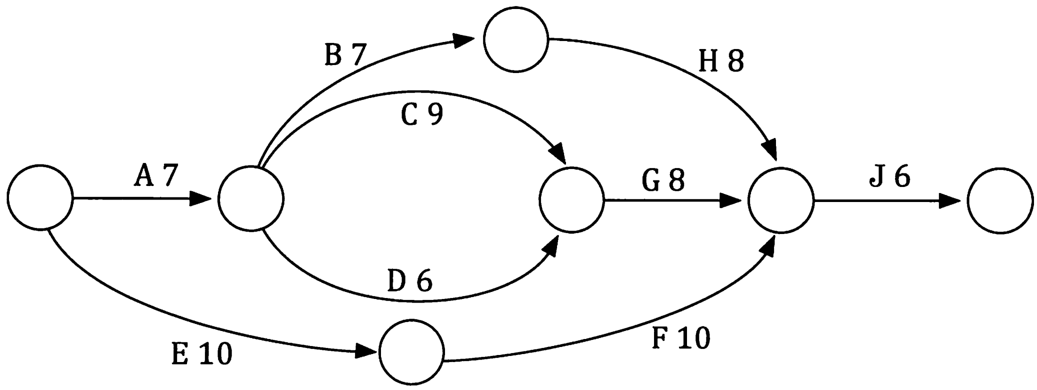
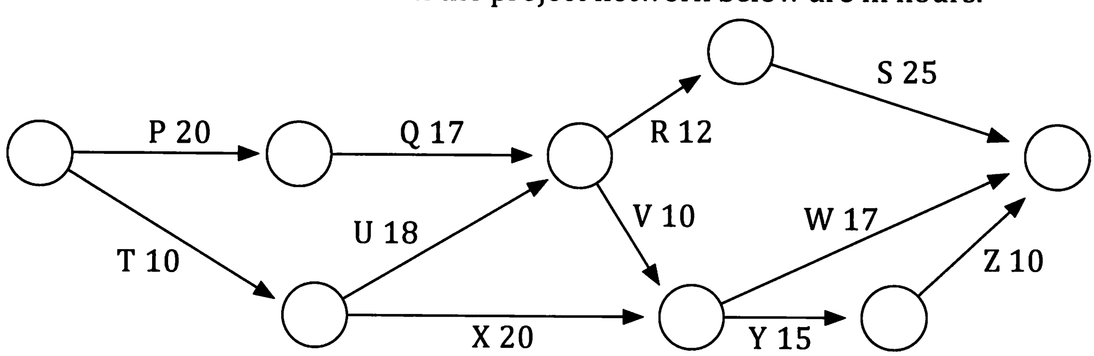
5. The times shown in the project network below are in days.



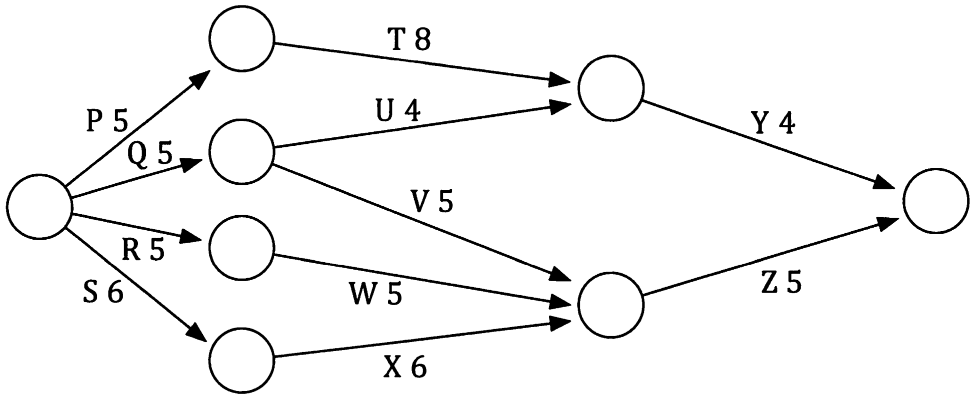

(a) Determine the minimum completion time.
(b) Determine the critical path.
(c) What is the earliest start time (EST) for task H?
(d) What is the latest start time (LST) for task H if the minimum completion time is to be unchanged from that given in the answer to part (a)?
(e) What is the float time for task H?
(f) What is the float time for task F?
(g) What is the float time for task D?
(h) Due to staff sickness it is necessary to reallocate the staff involved on tasks D and F. This will cause one of these jobs to require 5 extra days to complete. Which of the two tasks should be the one chosen to require this extra time if the completion time is to be kept as low as possible?

6. The times for tasks P to Z shown in the project network below are in hours.




(a) Determine the minimum completion time.
(b) Determine the critical path.
(c) What is the earliest start time after the commencement of the project that task X can commence?
(d) What is the latest start time after the commencement of the project that task X can commence without altering the minimum completion from (a)?
(e) What is the slack time for task X?
(f) What is the latest time after the commencement of the project that task W can commence without altering the minimum completion from (a)?
(g) What is the slack time for task W?

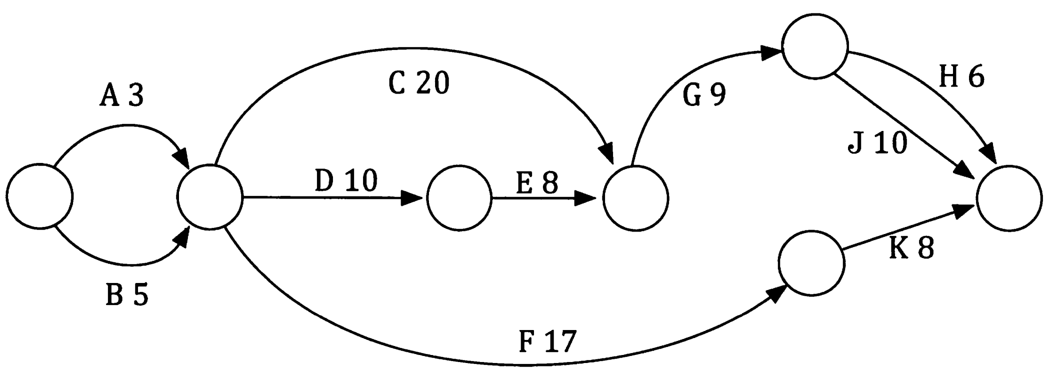
7. The times for tasks P to Z shown in the network below are in minutes.




(a) Determine the minimum completion time and critical path(s) for the above project network.

(b) How much slack time do activities Q and R have?

8. The times for the tasks shown in the project network below are in days.




(a) Determine the minimum completion time.

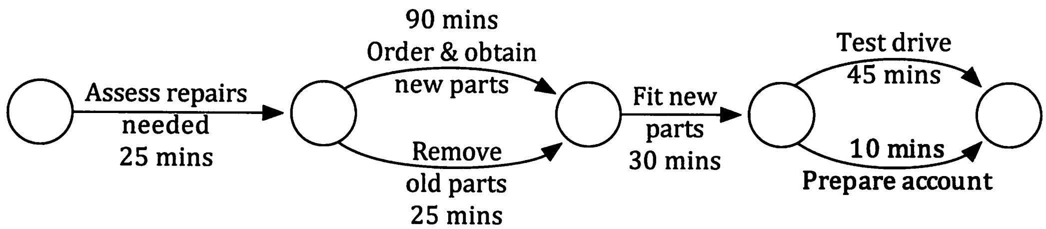
(b) Determine the critical path.

(c) What is the maximum number of days the commencement of task F could be delayed without altering the minimum completion time?

(d) What float time does activity E have?

(e) What float time does activity H have?

9. The project network below is for the repair of a car at a garage.

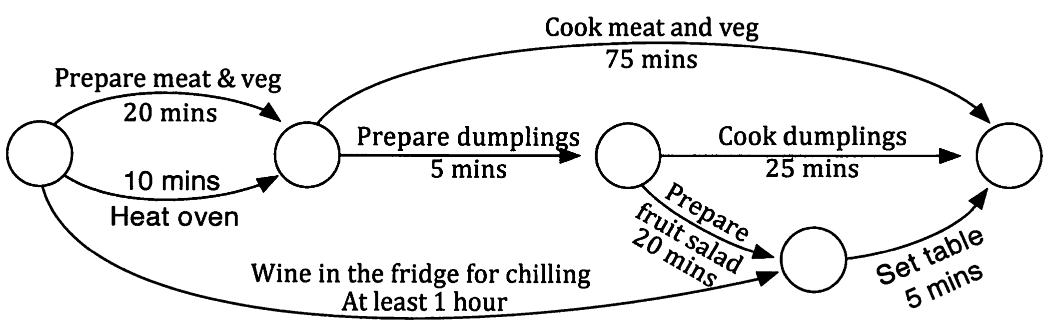



(a) Determine the minimum completion time.

(b) The owner of the car delivers it to the garage at 9 o'clock one morning. Determine the earliest time it will be ready, assuming the time for each task is as in the network.

(c) The new parts are in fact all in stock in the garage and do not need to be ordered from elsewhere. This task then takes just 5 minutes rather than the 90 allowed in the network. What is the earliest completion time now, given that work commences at 9 am?

10. A cook prepares a meal consisting of casserole, dumplings, fruit salad and chilled wine according to the project network shown below.




(a) If the meal has to be ready for 6 pm what is the latest time that the first activity can commence?

(b) What is the latest time that the warm wine can be placed in the fridge for chilling?

(c) With the wine in the fridge and just as she puts the meat and veg in the oven to cook and is about to start preparing the dumplings, the cook's phone rings. What is the maximum time she can spend on the phone without delaying the completion time? (Assume that she cannot prepare the dumplings and be on the phone at the same time.)

**Constructing a project network.**

The previous exercise involved questions being asked about given project networks. In some cases we may not be given the project network but instead be given sufficient information to allow the network to be constructed. Examples 2 and 3 are of this type.

**Example 2**

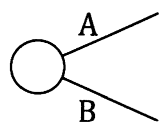
Construct a network for a project consisting of activities A to H listed below.

| Activity | Time |
|---|---|
| A | 7 hours |
| B | 9 hours |
| C | 8 hours |
| D | 3 hours |
| E | 7 hours |
| F | 4 hours |
| G | 5 hours |
| H | 3 hours |

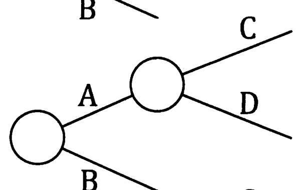
Order

- Activities A and B start together.
- Activities C and D can both commence when A is complete.
- Activities F and E can both commence provided both B and D have finished.
- Activity G can commence provided C and F are both complete.
- Activity H can commence provided E and G are both complete.

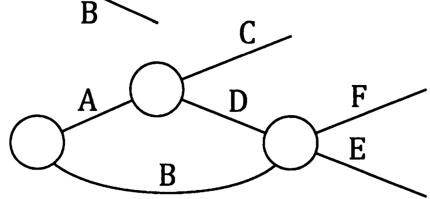
Activities A and B start together:



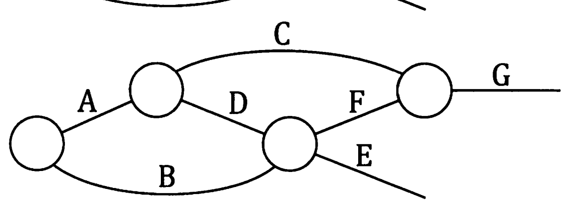

Activities C and D can both commence when A is complete:



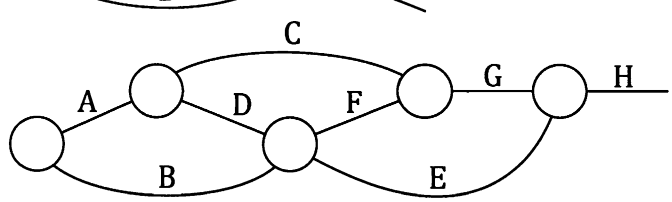

Activities F and E can both commence provided both B and D have finished:



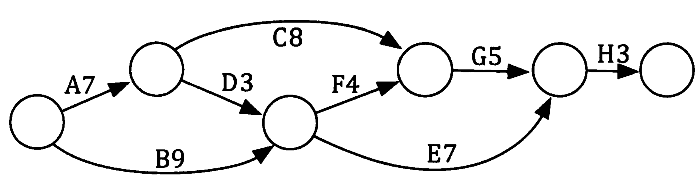

Activity G can commence provided C and F are both complete.




Activity H can commence provided E and G are both complete:




Complete the project network by adding the final circle, the arrows and the activity times:

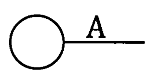


**Example 3**

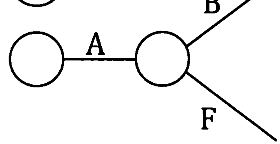
Construct a project network for a project that consists of the following tasks.

| Task | Time | Immediate Predecessors |
|---|---|---|
| A | 7 days | - |
| B | 4 days | A |
| C | 10 days | B |
| D | 3 days | B |
| E | 4 days | B |
| F | 6 days | A |
| G | 3 days | F |
| H | 7 days | F |
| I | 5 days | E, G |
| J | 4 days | D, H, I |

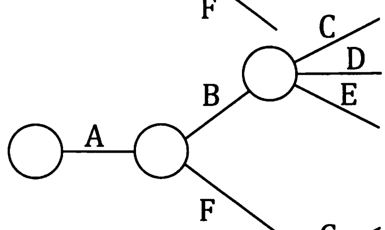
Start the network with those tasks that have no predecessors, in this case task A:



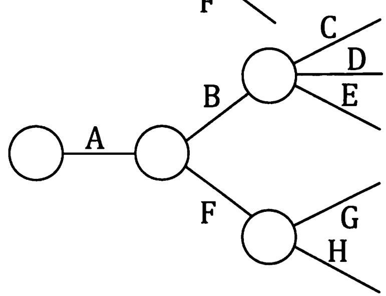

Tasks B and F have A as immediate predecessor:



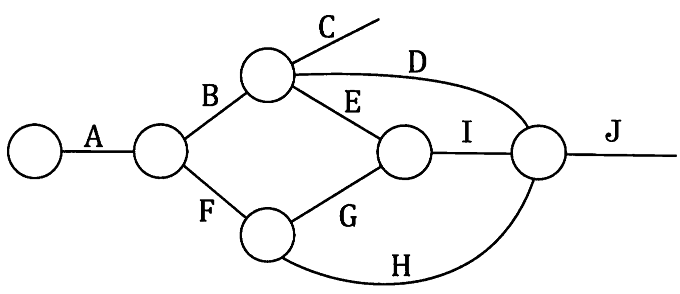

With B in place tasks C, D and E follow:



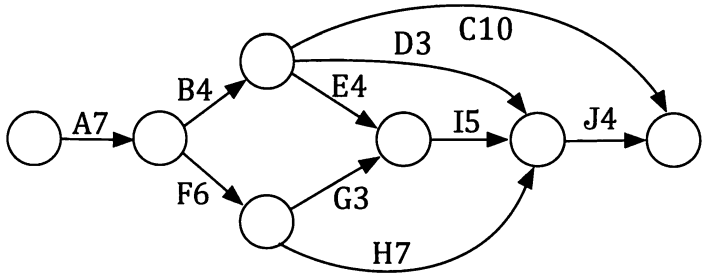

Tasks G and H can be placed :




The remaining tasks can be placed




and the network can be completed:



**Exercise 7B**

1. (a) Construct a project network for a project consisting of tasks P to X.

| Task | Scheduled Time | Immediate Predecessors |
|---|---|---|
| P | 7 weeks | – |
| Q | 5 weeks | – |
| R | 3 weeks | P |
| S | 4 weeks | P |
| T | 6 weeks | – |
| U | 9 weeks | T |
| V | 5 weeks | S, Q |
| W | 7 weeks | R |
| X | 2 weeks | V, W |

(b) Determine the minimum completion time and the critical path.

(c) How many weeks over the scheduled time can task U take without altering the minimum completion time? (Assuming all other tasks are completed as planned.)

2. (a) Construct a network for a project consisting of the following tasks.

| Task | Time | Immediate Predecessors |
|---|---|---|
| Q | 6 days | – |
| R | 8 days | – |
| S | 5 days | Q |
| T | 4 days | Q |
| U | 3 days | R |
| V | 7 days | R |
| W | 2 days | T, U |
| X | 4 days | S, W |
| Y | 3 days | V, Z |
| Z | 5 days | S, W |

(b) Determine the minimum completion time and the critical path.

(c) Task V is subject to delay and in fact takes 12 days to complete, not 7. What is the minimum completion time now?

3. (a) Construct a project network for a project that consists of the following tasks.

| Task | Time | Immediate Predecessors |
|---|---|---|
| A | 16 minutes | – |
| B | 21 minutes | A |
| C | 28 minutes | – |
| D | 25 minutes | B, C |
| E | 5 minutes | B, C |
| F | 25 minutes | E |
| G | 16 minutes | D, H |
| H | 17 minutes | E |

(b) Determine the minimum completion time and the critical path.

(c) By how many minutes can activity E be delayed without altering the minimum completion time?

4. (a) Construct a project network for a project consisting of activities A to G.

| Activities | Time | Order |
|---|---|---|
| A | 10 days | • Activities C and D start together. |
| B | 21 days | • Activities E and F must be after |
| C | 16 days | activity C. |
| D | 35 days | • Activities A and B need activity E to |
| E | 15 days | be finished before they can start. |
| F | 11 days | • Activity G can commence only after |
| G | 17 days | F is complete. |

(b) Determine the minimum completion time and the critical path.

(c) Lack of harmony on the industrial front threatens to delay task F. If all the other tasks can be completed according to the times given above, how many extra days can be allowed for activity F, over and above the scheduled 11, without increasing the minimum completion time?

5. (a) Construct a project network for a project consisting of activities A to I.

| Activities | Time | Order |
|---|---|---|
| A | 6 hours | • Activities A, D and H start together. |
| B | 7 hours | • Activity E must be after activity A. |
| C | 4 hours | • Activities B, F and I must be after D. |
| D | 7 hours | • Activity C can commence only after both |
| E | 10 hours | I and H are complete. |
| F | 3 hours | • Activity G can commence only after both |
| G | 5 hours | E and F are complete. |
| H | 9 hours | |
| I | 4 hours | |

(b) Determine the minimum completion time and the critical path.

(c) Which of the three tasks that start together, A, D and H, could experience delays causing them to be completed in a time greater than that stated above yet still not alter the minimum completion time for the project?

6. Construct a project network for a project consisting of tasks P to Z as listed below.

| Activity | Duration |
|---|---|
| P | 5 minutes |
| Q | 6 minutes |
| R | 10 minutes |
| S | 2 minutes |
| T | 4 minutes |
| U | 9 minutes |
| V | 4 minutes |
| W | 3 minutes |
| X | 10 minutes |
| Y | 4 minutes |
| Z | 4 minutes |

| Requirements |
|---|
| Q, T and W start when P is completed. |
| S can start when T and Y are completed. |
| X can start when W is completed. |
| R and Y can start when Q is completed. |
| Z can start when both S and X are completed. |
| U can start when both S and X are completed. |
| V can start when R and Z are completed. |

Determine the minimum completion time and the critical path.

7. (a) Construct a project network for a project that consists of tasks R to Z as listed below.

| Activity | Duration |
|---|---|
| R | 16 days |
| S | 22 days |
| T | 25 days |
| U | 20 days |
| V | 28 days |
| W | 17 days |
| X | 15 days |
| Y | 13 days |
| Z | 18 days |

| Order |
|---|
| R and T start together. |
| S, U, V start once both R and T are finished. |
| W starts when S is finished. |
| X starts when W is finished. |
| Y starts when U is finished. |
| Z starts when both V and Y are finished. |

(b) Determine the minimum completion time and the critical path.

(c) It is thought that activities U, V and Y may <u>each</u> require 2 extra days for completion than is given in the table. What would be the minimum completion time if all three of these tasks do require this extra time (and all other tasks have the duration shown in the table).

8. A company decides to interview three of the many applicants for a job. The three people are Mr James, Ms Kerry and Ms Maine.

The interview process will involve the following tasks:

| Activity | Description | Time (mins) |
|---|---|---|
| A | Three applicants to meet informally with interview panel for coffee. | 30 |
| B | Explain interview procedure to the applicants as a group. | 5 |
| C | Interview Mr James | 45 |
| D | Ms Kerry does written test | 30 |
| E | Ms Maine tours work site | 30 |
| F | Interview Ms Maine | 45 |
| G | Mr James does written test | 30 |
| H | Ms Kerry tours work site | 30 |
| I | Interview Ms Kerry | 45 |
| J | Ms Maine does written test | 30 |
| K | Mr James tours work site | 30 |
| L | Interview panel meet to make decision | 45 |
| M | Panel chairman informs applicants of panel's decision. | 2 |

The order in which these activities should be completed is given below.

- Activity A is the first activity and is followed by B.
- Activities C, D and E all follow B.
- Activities F, G and H occur once all of C, D and E have been completed.
- Activities I, J and K occur once all of F, G and H have been completed.
- Activity L follows I, J and K.
- Activity M follows L.

Display the process as a project network.

If activity A, the informal meeting over coffee, finishes at 9.30 am when does Ms Kerry finish her interview?

9. (a) Construct a project network for a project consisting of activities Q to Z.

| Activity | Time |
|---|---|
| Q | 4 hours |
| R | 4 hours |
| S | 7 hours |
| T | 9 hours |
| U | 3 hours |
| V | 11 hours |
| W | 14 hours |
| X | 7 hours |
| Y | 9 hours |
| Z | 3 hours |

Order
- Q, S and T can all start without needing other activities beforehand.
- R can commence provided Q has finished.
- U and V can commence provided R, S and T have all finished.
- W, X and Y can commence provided U has finished.
- Z can commence provided V, X and Y have all finished.

(b) Determine the minimum completion time and the critical path.

(c) Determine how many hours activity Z could run over its allocated time and still not alter the minimum completion time if
   (i) all the other tasks take exactly their allocated times,
   (ii) task Y is completed in 7 hours rather than the 9 allowed for.

10. A printing company has the job of printing a book for a client. Once they receive the original pages from the client they carry out the following activities:

| Activity | Description | Time (Days) |
|---|---|---|
| A | Obtain necessary materials (paper, card etc.). | 3 |
| B | Paste up originals to form masters. | 3 |
| C | Make printing plates from masters. | 1 |
| D | Print pages | 4 |
| E | Print cover | 1 |
| F | Fold and collate pages | 2 |
| G | Bind pages | 3 |
| H | Attach cover | 2 |
| I | Deliver to client | 1 |

The order in which these activities should be completed is given below.

- Obtaining necessary materials and pasting up can start together.
- Making the plates can only be done once the paste up is complete.
- The printing of the pages and the cover can both commence provided both the materials are obtained and the plates are made.

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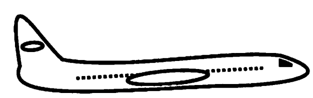
- The pages can be folded and collated as soon as they are all printed.
- Page binding can occur following folding and collation.
- The printed covers are attached once page binding is complete.
- When all complete the books are delivered to the client.

(a) Construct a project network for this job.
(b) Determine the minimum completion time and list the tasks that form the critical path.
(c) The covers are about to be printed when the machine that prints them breaks down and a new part has to be sent from Germany. What is the longest time this machine can be out of use without altering the minimum completion time?
(d) The company is considering purchasing a new machine that will fold, collate and bind the pages in a total of 4 days. By how much would this new machine reduce the minimum completion time?

11. An airline company is keen to reduce the time its planes are on the ground at a particular airport because of the high fees the airport authority charges. The company is charged for each minute that it has an aircraft at an airport terminal.



Construct a project network for the "on ground" tasks listed below and determine the minimum completion time.

| Task | Description | Time (Mins) | Preceded by |
|---|---|---|---|
| A | Support service vehicles in position | 5 | — |
| B | Passengers disembark. | 15 | A |
| C | Luggage off. | 8 | A |
| D | Old food trolleys off. | 5 | A |
| E | Refuel | 10 | A |
| F | Security check of empty aircraft | 15 | B, C |
| G | New passengers on | 25 | F |
| H | Luggage on | 12 | F |
| I | Food trolleys on | 5 | D |
| J | Remove old waste disposal units | 8 | A |
| K | Fit new waste disposal units | 5 | J |

Which tasks should they attempt to speed up if they are to reduce the time at the terminal?

12. Though generally avoided in this text, the following situation is one requiring the introduction of a "dummy activity" of zero weight for a network diagram to be formed. Use this idea to represent the following as a project network with dummy activity Q, and determine the critical path and shortest completion time.

Tasks A (5 hours) and B (4 hours) start together.
Task C (7 hours) can commence when both A and B are completed.
Task D (6 hours) can commence when task A is completed.

13. In order to keep the production of a school year book on time the organising committee list the tasks involved and the order in which they must occur.

| TASK | ACTIVITY | TIME (weeks) |
|---|---|---|
| A | Advertise fact that articles are required and receive articles. | 10 |
| B | Take individual photographs of all year twelve students. | 3 |
| C | Check all individual photographs and retake as necessary. | 1 |
| D | Get personal details such as nickname etc. from each yr 12. | 3 |
| E | Have personal details typed to same style and format. | 1 |
| F | Select articles to be included from those submitted. | 1 |
| G | Edit selected articles for grammar and acceptability. | 1 |
| H | Have all selected articles typed to same style and format. | 1 |
| I | Paste up articles, personal details and photos as ordered pages. | 3 |
| J | Print year book. | 2 |

Order.

Tasks A, B and D start together.
Task C can commence when task B finishes.
Task E can commence when task D finishes.
Task F can commence when task A finishes.
Task G can commence when task F finishes.
Task H can commence when task G finishes.
Task I can commence when tasks C, E and H are finished.
Task J can commence when all other tasks have been completed.

Represent this information as a project network and determine the minimum completion time.

**Miscellaneous Exercise Seven.**

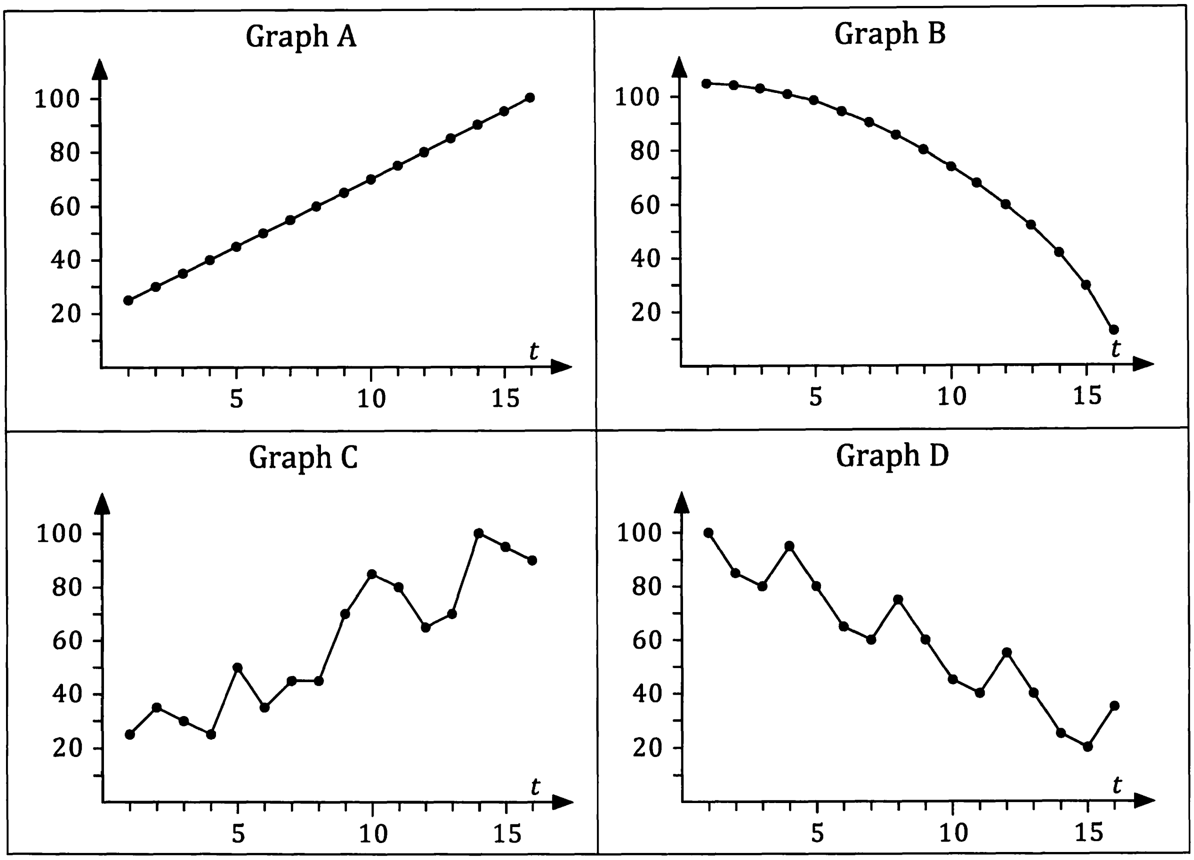
**This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.**

1. Explain what it means to say that the seasonal index for January is 1·12.

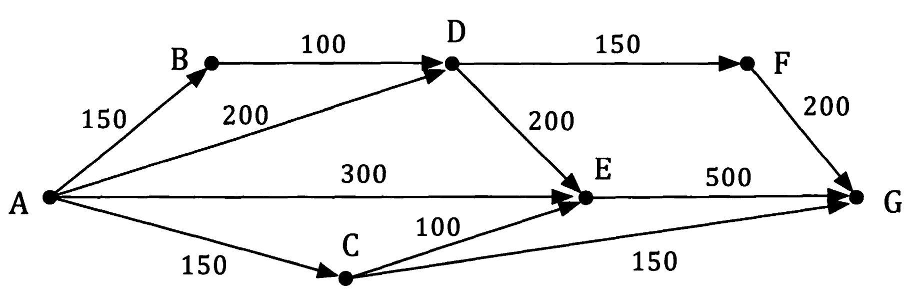
2. A loan for \$280000 is taken out with compound interest charged at a rate of 10% per annum compounded monthly. If repayments of \$2800 are made at intervals of one month after the start of the loan, how much is owed five years after its commencement?

3. An analysis of the number of cars a particular manufacturer sells in each month, over a number of years, gives the seasonal index for January as 92% and for February as 88%.
The following year the manufacturer sells 2031 cars in January and 1997 cars in February. Express the number of cars sold in these two months as seasonally adjusted numbers.

4. Which of the following graphs show
   (a) an increasing trend,
   (b) a decreasing trend,
   (c) data suitable for linear modelling,
   (d) data with irregular fluctuations,
   (e) data with seasonal fluctuations?

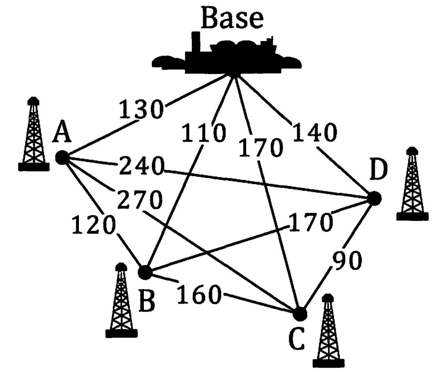



5. Determine the maximum number of units that can be delivered at the sink, G, from the source, A, for the network shown below.




If the capacity of arc AC could be increased from the 150 units shown in the diagram to 250 units how would this alter the maximum number of units that can be delivered from A to G?

6. The network on the right shows four proposed off-shore drilling wells and the land based processing base. Pipelines are to be laid connecting each well to the base, either directly or via one or more of the other wells. The cost of each line is shown in \$1000s.
Determine the network of pipes required to complete the desired connections whilst keeping the total cost to a minimum, and find this minimum cost.




7. Following the large number of fires that have occurred in a particular state forest over recent years it is decided that five lookout positions need to be established in the area. A system of roads is to be developed so that each of the lookout positions is connected to every other lookout position, either directly or indirectly, by road. The road distances between each pair of lookout positions would be as given in the table below, in kilometres.

| | Lookout 1 | Lookout 2 | Lookout 3 | Lookout 4 | Lookout 5 |
|---|---|---|---|---|---|
| Lookout 1 | – | 5·0 | 10·5 | 11·2 | 6·0 |
| Lookout 2 | 5·0 | – | 7·0 | 10·0 | 9·0 |
| Lookout 3 | 10·5 | 7·0 | – | 5·5 | 10·8 |
| Lookout 4 | 11·2 | 10·0 | 5·5 | – | 7·8 |
| Lookout 5 | 6·0 | 9·0 | 10·8 | 7·8 | – |

Which pairs of lookout positions should have direct road connections constructed if the total length of construction is to be kept to a minimum?

8. The network below shows a system of roads linking town A to town Z.
Points B, C, D, E and F are road junctions.
The number on each road gives the maximum number of vehicles (in hundreds) that can travel along that section of road, in the direction indicated, per hour.
Determine the greatest number of vehicles that could arrive at Z per hour, having come from A.